

RAG Knowledge Base + Chatbot

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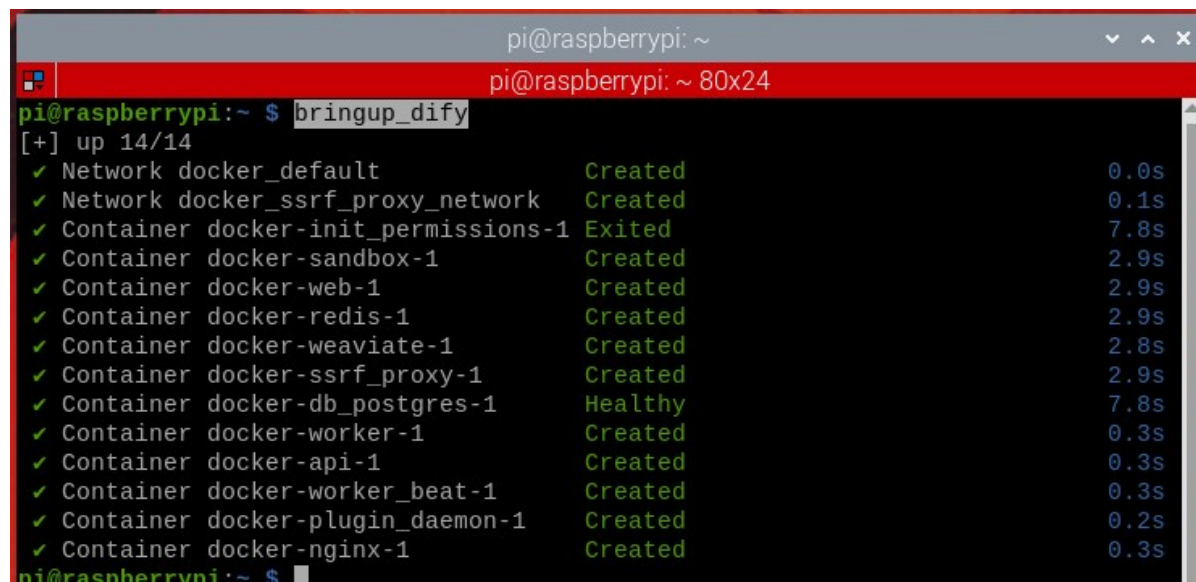
1. Course Content

- Based on the previous RAG knowledge base and chatbot, develop an AI application that combines the two to improve the answering capabilities of large AI models in specific domains, reducing AI model hallucinations through the RAG knowledge base.

2. Starting the Dify Service

- Connect to the vehicle's computer via VNC or SSH, and enter the following command in the terminal:

```
bringup_dify
```



```
pi@raspberrypi: ~  
pi@raspberrypi: ~ 80x24  
pi@raspberrypi:~ $ bringup_dify  
[+] up 14/14  
✓ Network docker_default Created 0.0s  
✓ Network docker_ssrf_proxy_network Created 0.1s  
✓ Container docker-init_permissions-1 Exited 7.8s  
✓ Container docker-sandbox-1 Created 2.9s  
✓ Container docker-web-1 Created 2.9s  
✓ Container docker-redis-1 Created 2.9s  
✓ Container docker-weaviate-1 Created 2.8s  
✓ Container docker_ssrf_proxy-1 Created 2.9s  
✓ Container docker-db_postgres-1 Healthy 7.8s  
✓ Container docker-worker-1 Created 0.3s  
✓ Container docker-api-1 Created 0.3s  
✓ Container docker-worker_beat-1 Created 0.3s  
✓ Container docker-plugin_daemon-1 Created 0.2s  
✓ Container docker-nginx-1 Created 0.3s  
pi@raspberrypi:~ $
```

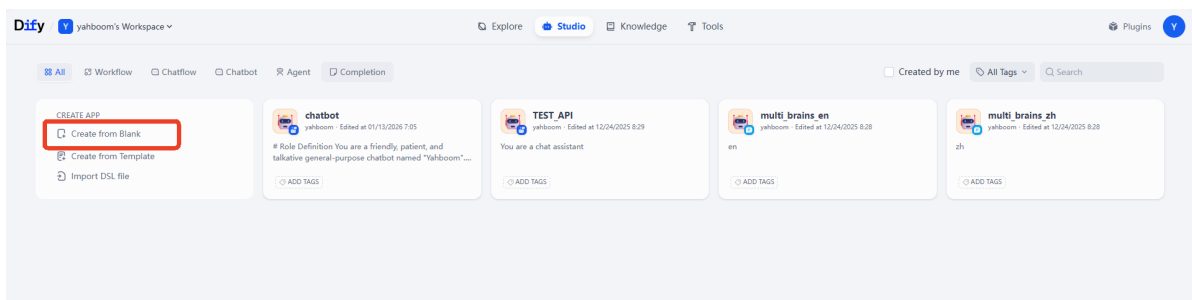
- Check the vehicle's IP address (you can view it on the OLED screen, using `ifconfig`, or directly in the terminal). Enter the vehicle's IP address directly in the browser's address bar to access the Dify management page.

```
jetson@yahboom: ~
jetson@yahboom: ~ 80x24
[System Information]
-----
IP_Address_1: 192.168.2.49
IP_Address_2: 192.168.12.34
MACHINE: OrinNx | ROS_DISTRO: humble | ROS_DOMAIN_ID: 30
ROBOT_TYPE: ROSMASTER-M3Pro | CAMERA_TYPE: dabai_dcw2 | RADAR: Tmini-plus*2
jetson@yahboom:~$
```

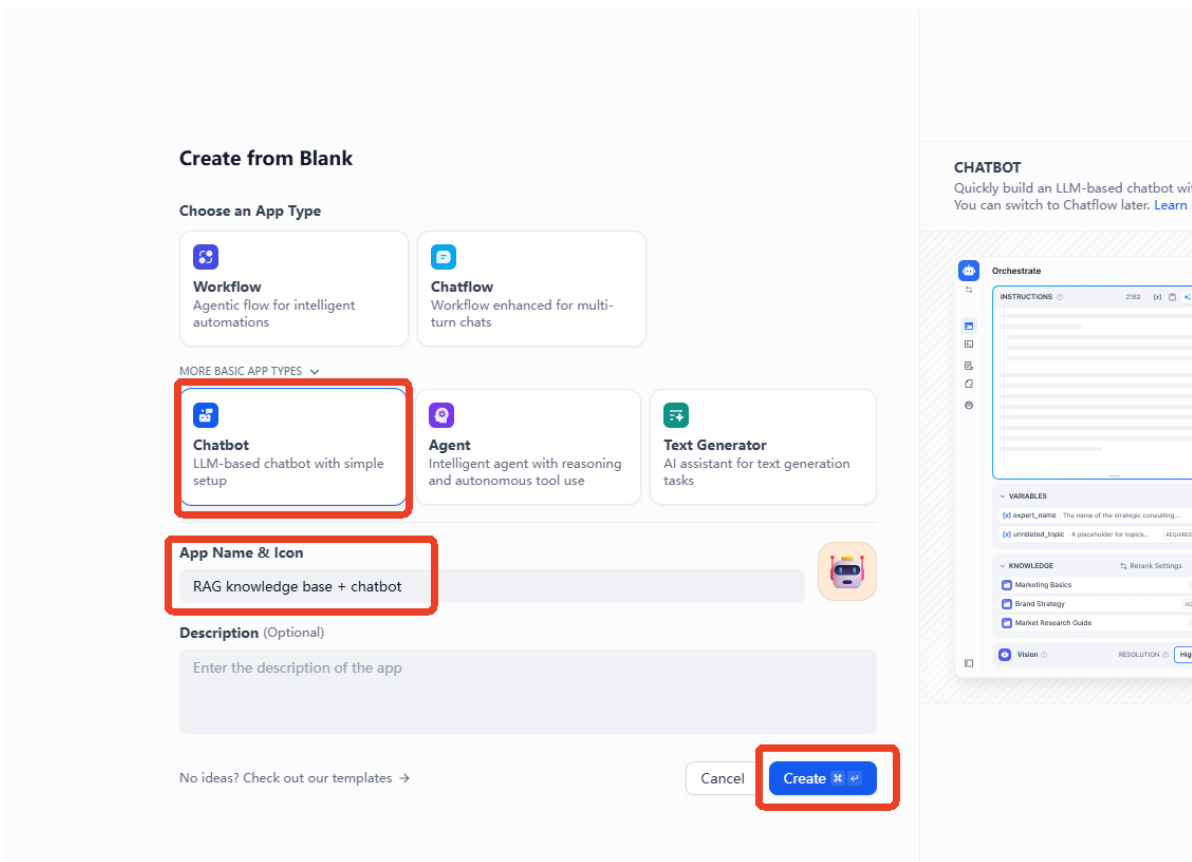
3. Case Study 1: Task Planning

[!TIP]

- Case Study 1 simulates the functionality of the ROSMASTER-M3decision-making layer.
- On the homepage, click "Create from Blank".



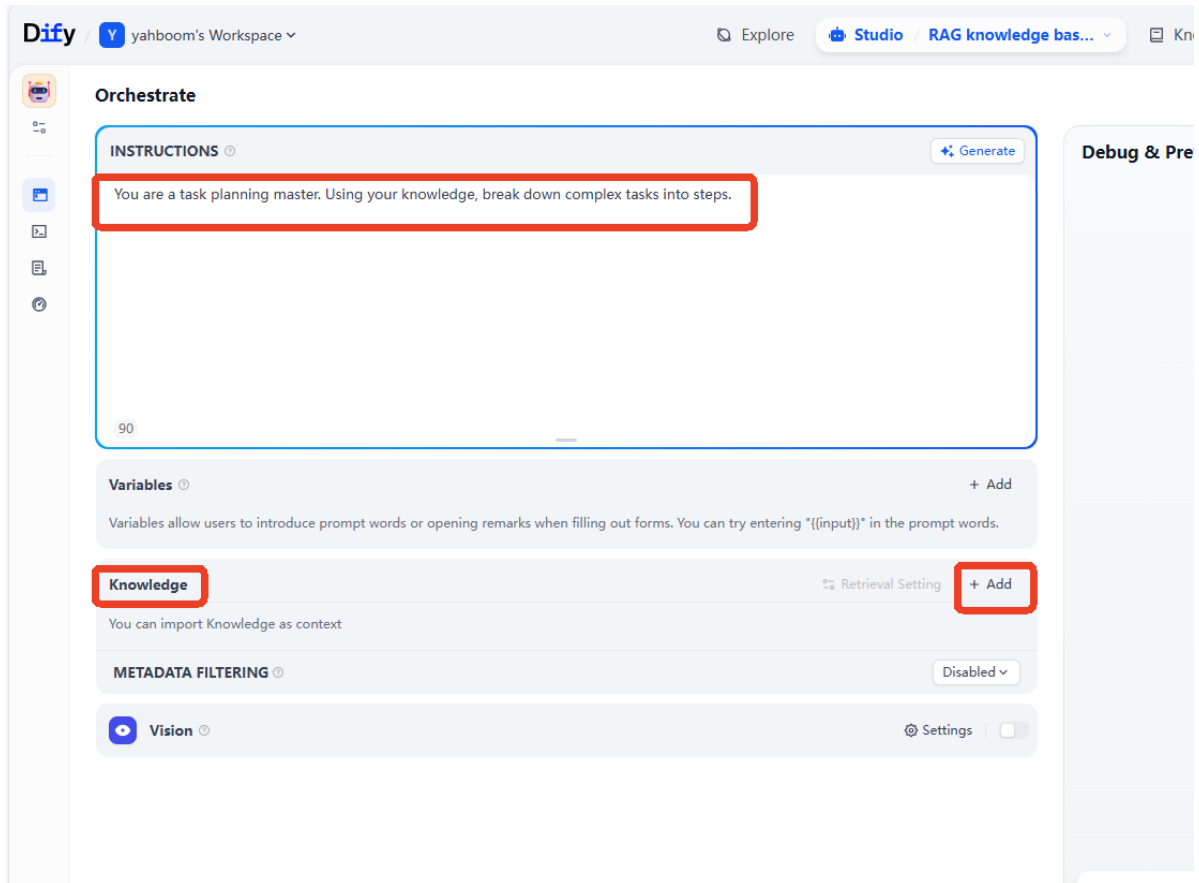
- Click to select "Chat Assistant" under Chatbot ->App Name & Icon -> Create.



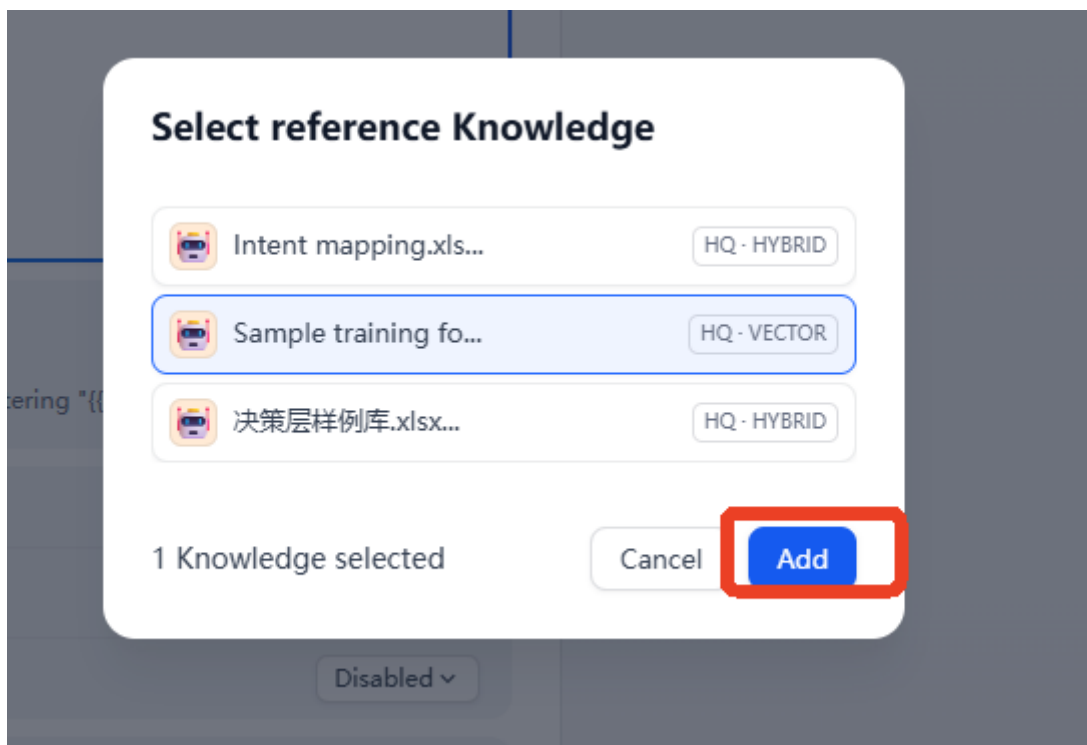
- Example prompt:

You are a task planning master. Using your knowledge, break down complex tasks into steps.

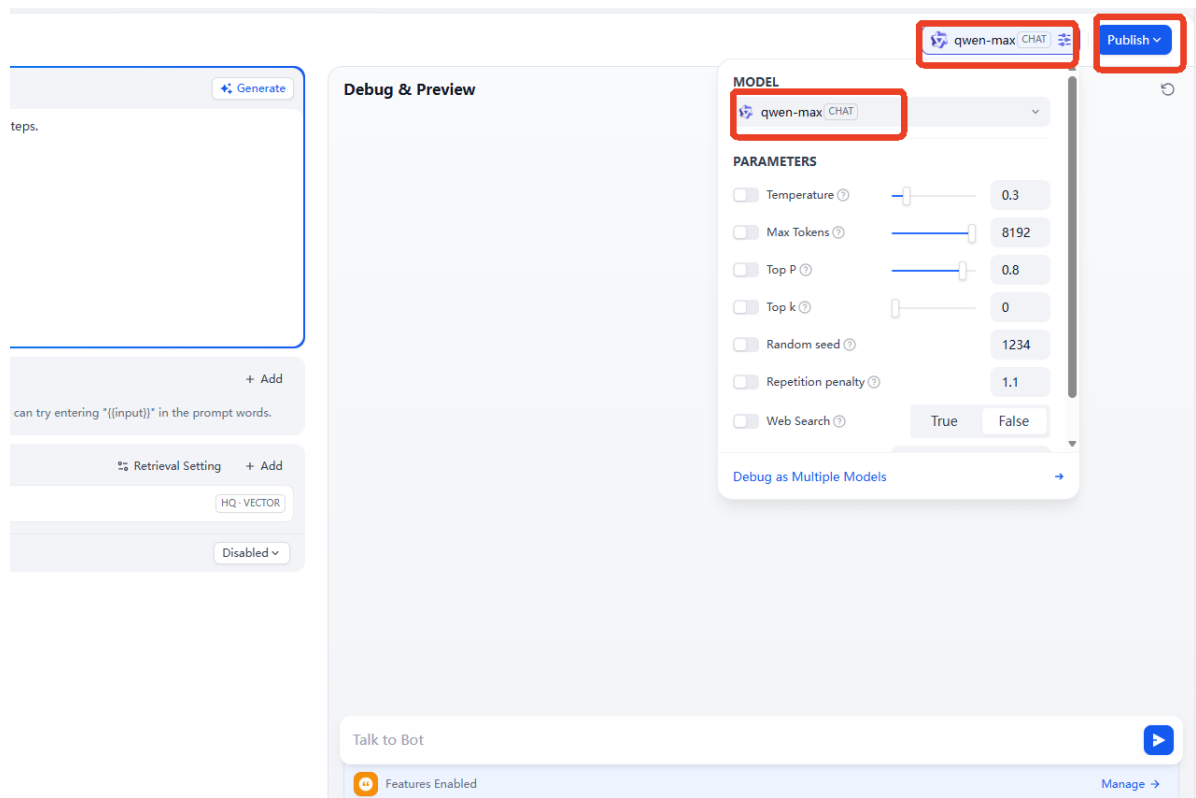
- After creating the application, fill in the prompt in the "INSTRUCTIONS", and then click "Add" in the knowledge.



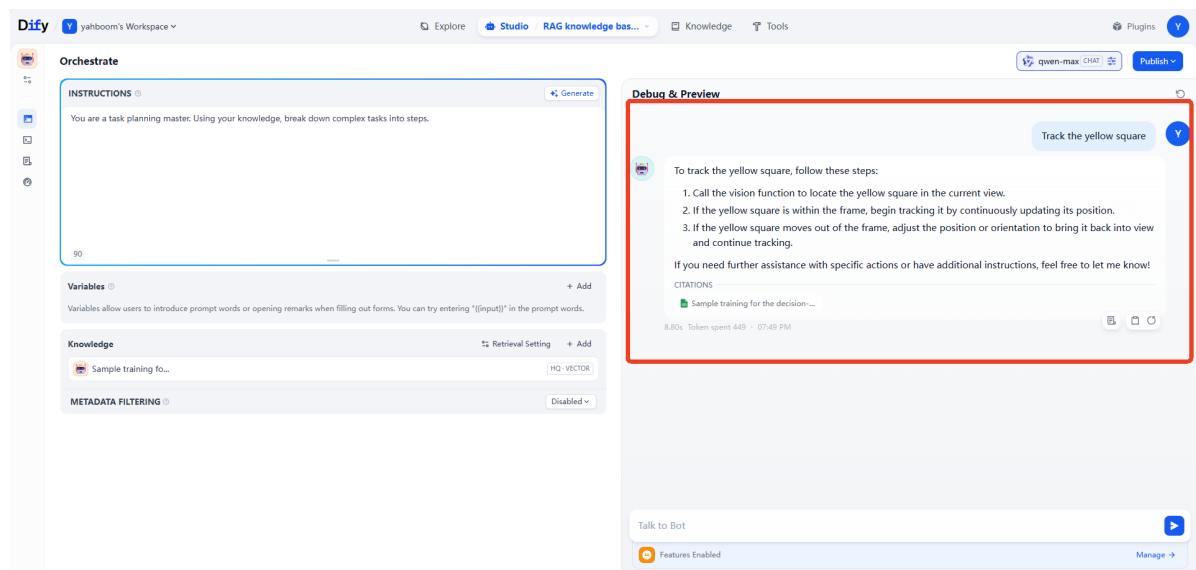
Select the knowledge base you want to add, then click Add.



- Then select an AI large language model to connect to. Here, we use the `qwen-max` model as an example.



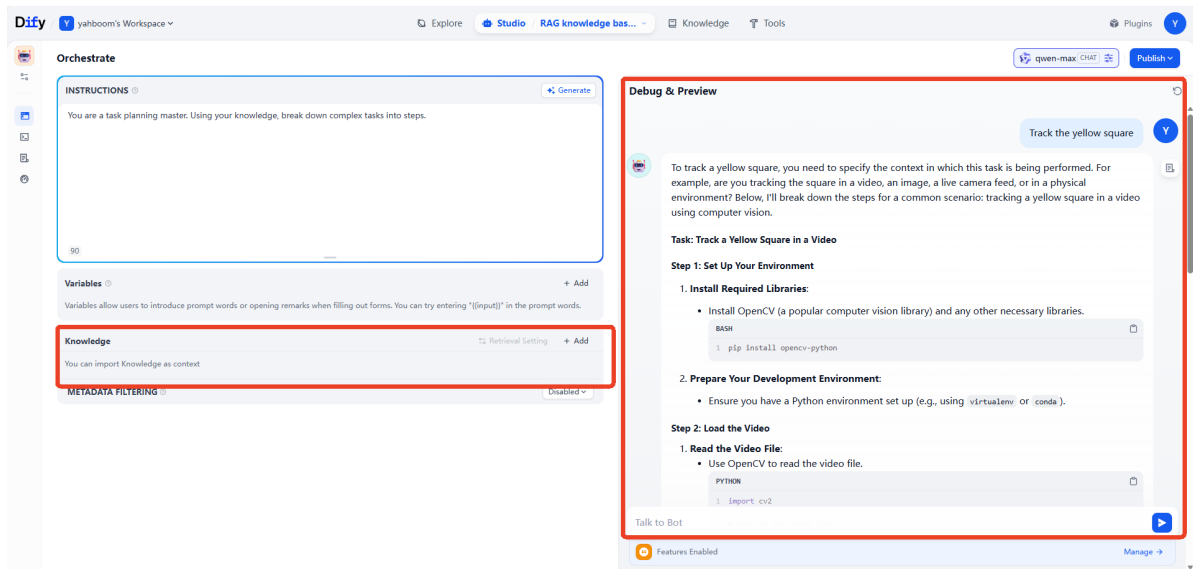
- After that, enter the test dialogue content. Here, we use removing machine code as an example. You can see that the AI large language model can think and answer based on the content of the knowledge base. By adding training examples to the knowledge base in this way, you can quickly expand the AI large language model's response capabilities and learning abilities in different task scenarios and fields.



- We can first delete the knowledge base to compare the AI large language model's response without the RAG knowledge base. You can see that without a knowledge base, the general-purpose AI model from the model provider cannot handle tasks in vertical domains.

[!TIP]

- This practical example can be understood and mastered in conjunction with the theoretical knowledge of **2. RAG Retrieval Augmentation and Training Examples**.

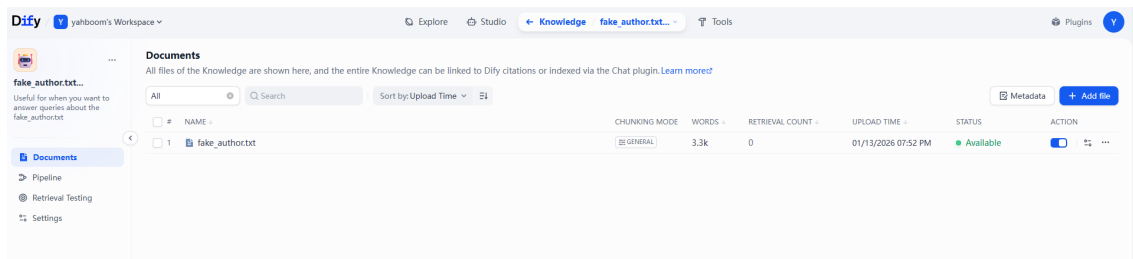


4. Case Study 2: Knowledge Management

- When we have internal documents that need to be managed by a large language model, we can use the AI large language model combined with RAG and a knowledge base to allow the large language model to respond based on actual business documents. This is very helpful for vertical fields such as finance and healthcare.

4.1 Creating a Knowledge Base

- Here, we use a sample document as an example, which contains a fictional author and a collection of fictional works. The sample document can be found in the folder for this section of the course.



4.2 Creating an Application

- Example Prompt

Role Positioning

Born in 1988 in a southern coastal city, he worked as an advertising copywriter for 5 years before resigning in 2015 to write full-time. He excels at blending delicate emotional portrayals with light science fiction elements, exploring the loneliness and connection of modern urban dwellers, and the balance between technological development and human warmth. His works combine a healing quality with thought-provoking themes, targeting primarily urban youth aged 25-40.

- Fill in the example prompt and select the virtual author's knowledge base. Then, ask questions about the works in the material. At this point, the AI large language model can answer using the private knowledge base.

Dify

yahboom's Workspace

Explore

Studio

RAG knowledge bas...

Knowledge

Tools

Plugins

Y

RAG knowledge base + C...

CHATBOT

Orchestrate

API Access

Logs & Annotations

Monitoring

INSTRUCTIONS

Generate

Role Positioning

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489

Variables

Add

Variables allow users to introduce prompt words or opening remarks when filling out forms. You can try entering "{input}" in the prompt words.

Knowledge

Retrieval Setting

Add

fake_author.txt

HQ HYBRD

METADATA FILTERING

Disabled

Debug & Preview

qwen-max

CHAT

Publish

Introduce the work 'A Glimmer of Light in the Neon Depths'

Y

'A Glimmer of Light in the Neon Depths' is an urban romance novel published in 2020. The story revolves around Su Wan, an advertising planner facing workplace challenges and emotional turmoil in a bustling first-tier city, and Chen Yu, an AI sound engineer. Their encounter and growing commitment form the heart of the narrative.

The book delves into the theme of 'emotional connection in an era of loneliness,' exploring how two individuals find mutual redemption in a neon-lit urban environment. It vividly portrays the survival pressures faced by urban white-collar workers, the alienation in their interpersonal relationships, and the process of building trust against the backdrop of rapid technological advancements.

Upon its release, "A Glimmer of Light in the Neon Depths" topped the urban literature bestseller list for three consecutive months. Readers praised it as "a healing work that best understands the loneliness of contemporary young people."

CITATIONS

fake_author.txt

9.08s Token spent 722 - 07:56 PM

Talk to Bot

Features Enabled

Manage